RHU LabChain



Block-Chain Powered Laboratory Records and Verification System

RHU of M’lang, North Cotabato

Problem statement

- Paper records complicate retrieval and tracking.
- Released reports need reliable integrity checks.
- Sensitive findings require controlled access.
- Replaced reports must remain clearly identifiable.

Research gap

- Digitization alone does not establish report integrity.
- Laboratory workflows need evidence linked to releases.
- Report versions need traceable verification histories.
- This study integrates these functions for RHU workflows.

Objectives

- Develop the laboratory and blockchain verification prototype.
- Evaluate access controls and report-integrity verification.
- Assess usability, accessibility and perceived workflow efficiency.
- Examine user acceptance, trust and adoption intentions.

Conceptual framework

Input: laboratory records and approved reports

Validate: identifiers, required fields and associations

Fingerprint: PDF digest and event commitment

Anchor: record the commitment in Fabric

Verify: compare documents, lifecycle and peers

Dataset and participants

- Synthetic records and sample laboratory reports
- Patient survey: 30 respondents; convenience sampling
- Staff survey: 11 personnel; total enumeration
- Staff: eight laboratory and three IT personnel
- Separate questionnaires: 30 items; five-point agreement scale

Laboratory system process

Register patient → create laboratory order

Handle specimen → enter laboratory findings

Verify results → prepare report snapshot

Sign and approve → release official PDF

Capture event → deliver blockchain commitment

Preprocessing and fingerprints

Validate identifiers, fields and record associations.

D = SHA-256(exact released PDF bytes)

Canonical event binds D, version and lifecycle.

C = SHA-256(canonical event bytes)

Fabric stores C with selected anchor metadata.

Why Hyperledger Fabric?

Permissioned identities govern transaction participation.

Chaincode validates and preserves report anchors.

Org1: peer1 + peer2 —— **Org2: peer3 + peer4**

One VPS; one orderer; shared failure risk.

Blockchain commitment process

- Capture release and outbox together
- Worker proposes; required organizations endorse
- Orderer sequences; peers validate and commit
- Worker checks committed evidence before confirmation
- Pending delivery does not mean blockchain confirmation

Verification and report lifecycle

Scan QR → locate the registered report

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Check current, revoked or superseded status

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Compare submitted PDF with released PDF digest

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Query four peers for required report events

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Four fresh matching observations support full verification

Evaluation and observed results

- Original PDF → MATCH in documented example
- Changed PDF → MISMATCH in documented example
- Four-peer agreement → VERIFIED in documented example
- One peer offline → PARTIAL, three peers verified
- Screenshots document scenarios, not measured performance rates

Interpreting verification results

MATCH: compared PDF bytes match issued artifact

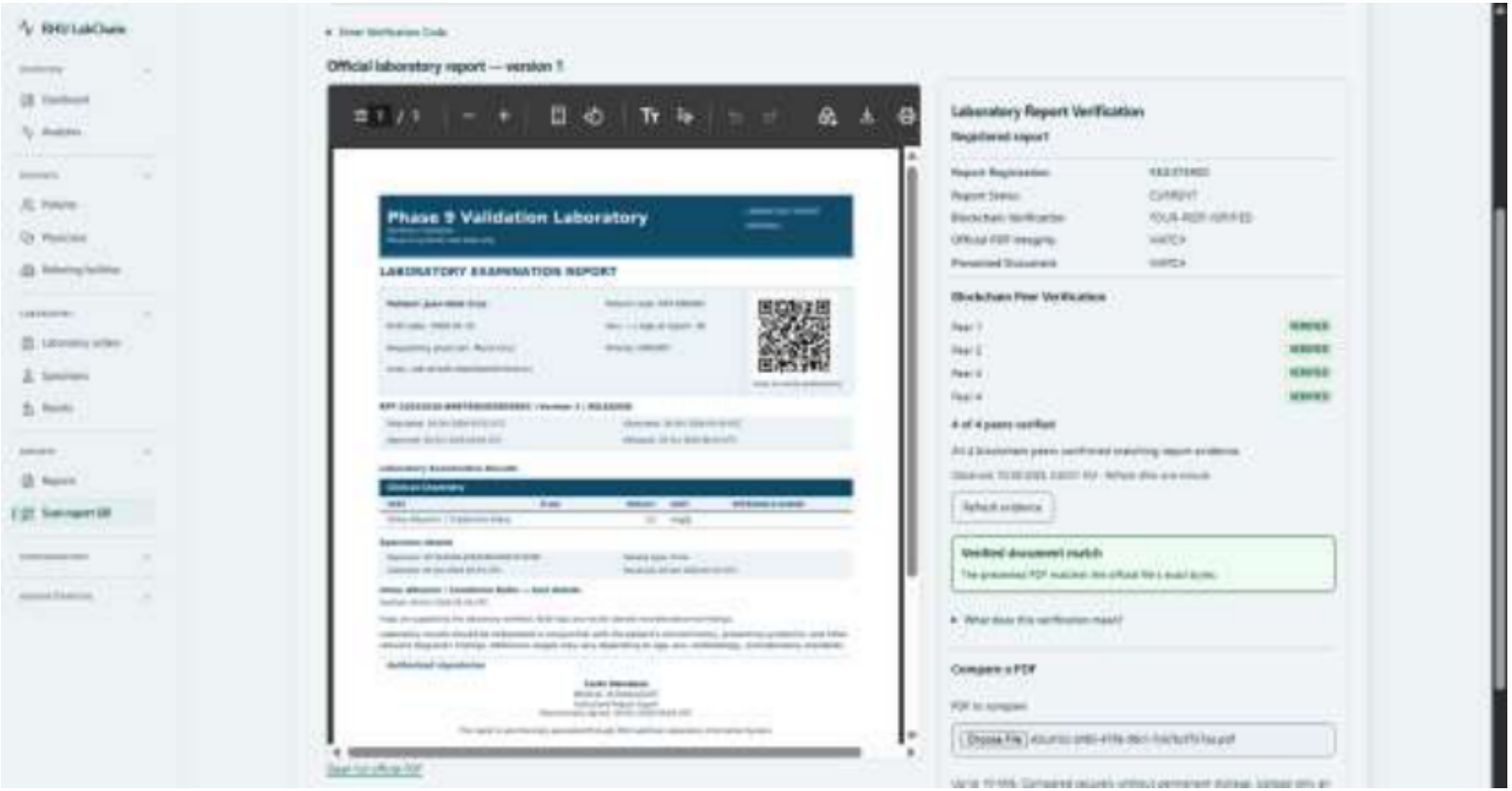
MISMATCH: bytes differ; cause remains undetermined

PARTIAL: evidence is incomplete across required peers

CONFLICT: contradictory evidence blocks full verification

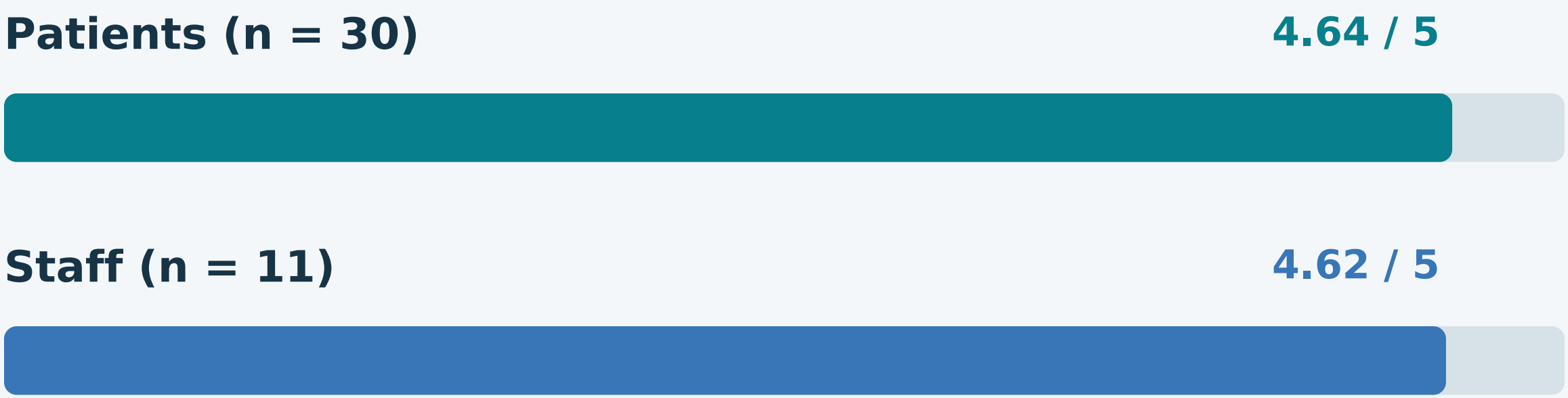
QR recognition alone cannot verify printed values

System output



Official PDF, lifecycle and four-peer verification

Overall survey means



Favorable perceptions; measured efficiency gains remain unproven.

Study limitations

- Shared hosting limits independent fault tolerance.
- PDF integrity does not establish clinical accuracy.
- Screenshots do not establish repeated performance benchmarks.
- Patient convenience sampling limits broader generalization.
- Aggregate tallies prevent respondent-level reliability analysis.

Recommendations

- Distribute peers and orderers across independent hosts.
- Test conflicting evidence and outage recovery.
- Document role and ownership access-denial checks.
- Measure task time against paper-based workflows.
- Retain anonymized respondent-level data for reliability analysis.

Conclusion and contribution

Controlled laboratory workflows.

Traceable report versions.

Blockchain evidence that can be checked.

Academic prototype; further clinical evaluation remains necessary.